



Overview of SDSC Computing Resources

*Expanse, CloudBank, TSCC, Voyager, PNRP,
and Cosmos*

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**SAN DIEGO
SUPERCOMPUTER
CENTER**

UC San Diego

HALICIOĞLU SCHOOL OF DATA SCIENCE
AND COMPUTING

SDSC Computing Resources

- SDSC operates and supports users on several NSF ACCESS/NAIRR allocated resources
- NSF resources are in two categories:
 - **Category I systems** – large-scale, production-ready computational resources that serve as capacity resources. At SDSC Expanse and CloudBank fit under this category
 - **Category II systems** - Innovative Prototypes/Testbeds for advanced computing and data analysis. These resources are experimental systems that deploy novel technologies, architectures, and usage models. At SDSC Voyager, PNRP, and Cosmos fit under this category.
- Category II systems typically have two phases: 1) the initial 3-year testbed phase where there are focused long term engagements with research groups to evaluate the new technologies and develop usage models, and 2) the 2-year allocated phase where systems are open to all users under the ACCESS or NAIRR Pilot programs
- Currently **Voyager and PNRP are in the allocations phase** and **Cosmos is in the testbed phase**
- In addition to the NSF resources, SDSC operates the Triton Shared Computing Cluster (TSCC)

EXPANSE

NSF Funded Category I System

NSF Award # 1928224

EXPANSE⁷ COMPUTING WITHOUT BOUNDARIES

5 PETAFLOP/S HPC and DATA RESOURCE

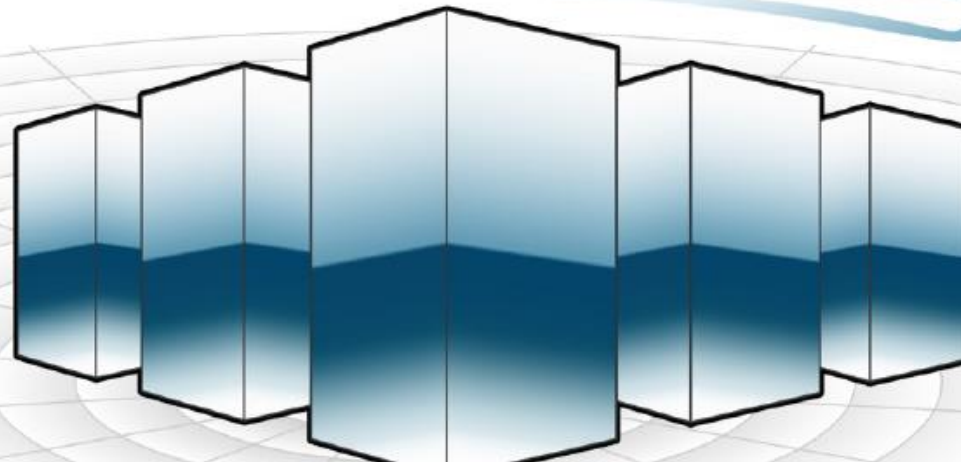
17

HPC RESOURCE

~~13~~ Scalable Compute Units
728 Standard Compute Nodes
52 GPU Nodes: 208 GPUs
4 Large Memory Nodes

DATA CENTRIC ARCHITECTURE

12PB Perf. Storage: 140GB/s, 200k IOPS
Fast I/O Node-Local NVMe Storage
7PB Ceph Object Storage
High-Performance R&E Networking



REMOTE CI INTEGRATION

CLOUD



Heterogeneous Resources



Open Science Grid



LONG-TAIL SCIENCE

Multi-Messenger Astronomy
Genomics
Earth Science
Social Science

INNOVATIVE OPERATIONS

Composable Systems
High-Throughput Computing
Science Gateways
Interactive Computing
Containerized Computing
Cloud Bursting

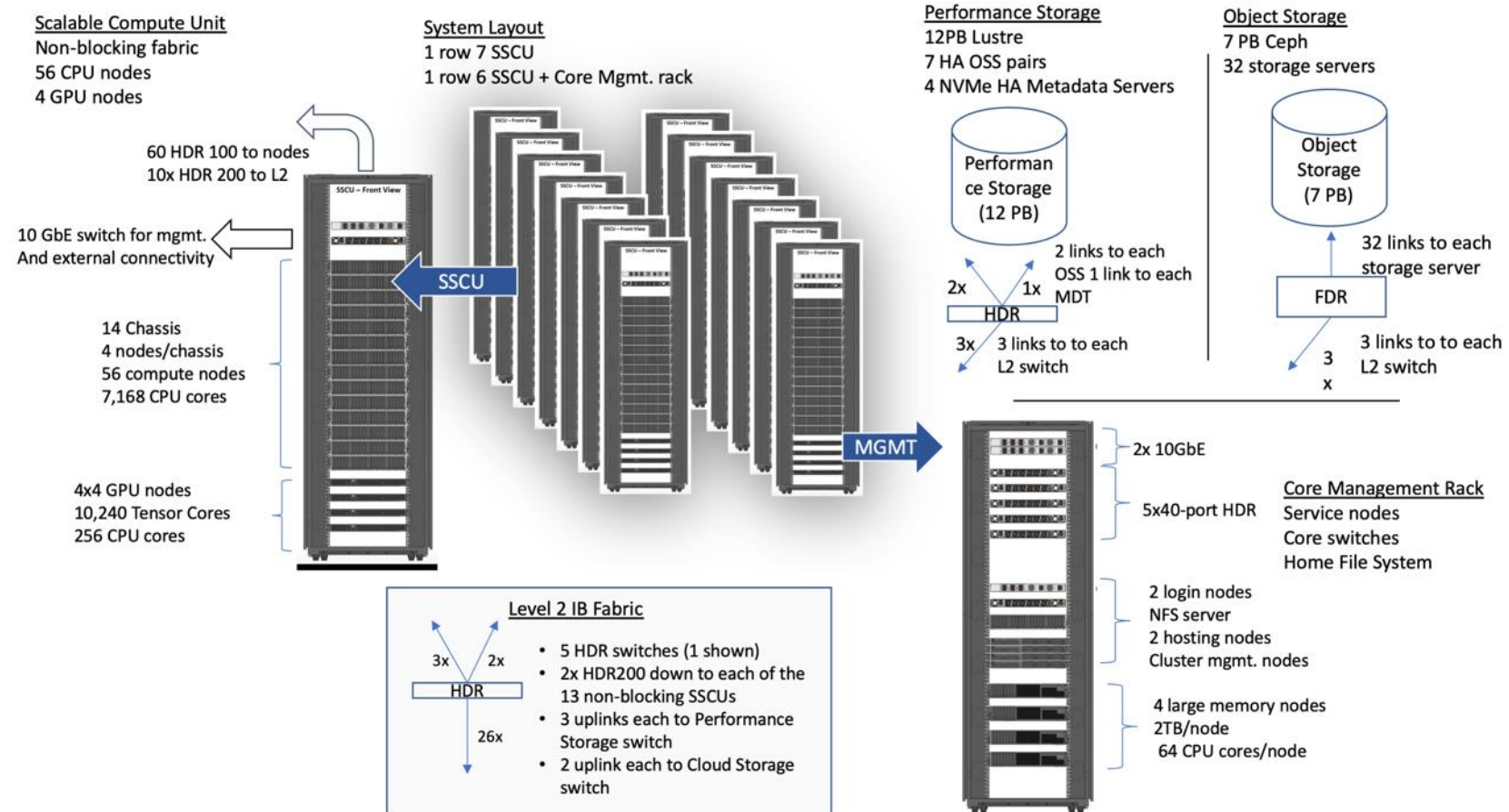
NSF Award # 1928224

PIs: Mike Norman (PI), Ilkay Altintas, Amit Majumdar, Mahidhar Tatineni, Shawn Strande

Expanse is a heterogeneous architecture designed for high performance, reliability, flexibility, and productivity

System Summary

- 14 SDSC Scalable Compute Units (SSCU)
- 784 x 2s Standard Compute Nodes
- 100,352 Compute Cores
- 200 TB DDR4 Memory
- 56x 4-way GPU Nodes w/NVLINK
- 224 V100s
- 4x 2TB Large Memory Nodes
- HDR 100 non-blocking Fabric
- 12 PB Lustre High Performance Storage
- 7 PB Ceph Object Storage
- 1.2 PB on-node NVMe
- Dell EMC PowerEdge
- Direct Liquid Cooled



Growth of Expanse

- ***Original Configuration (13 racks): Category 1: Capacity System, NSF Award # 1928224***
 - 728, 2-socket AMD Rome-based compute nodes (2.25 GHz EPYC; 64-core/socket). 93,184 compute cores in total.
 - 52 4-way GPU nodes with V100 GPUs w/NVLINK
- ***Partnership to Advanced Throughput Computing (PATH) racks (2) – NSF funded, 2022:***
 - 112 2-socket AMD Milan-based compute nodes; 512 GB of memory per node
 - 8 4-way GPU nodes w/ A100 GPUs
- ***Industry rack (funded by UCSD/SDSC):***
 - 56 2-socket AMD Rome-based compute nodes
 - 4 4-way GPU nodes based on V100 w/NVLINK
- ***CW3E rack, 2023; Moved in 2025 to AWARE cluster***
 - 60 2-socket AMD Milan-based compute nodes; 256 GB of memory per node
- ***Expanse AI Resource (NAIRR Pilot supplement), 2025***
 - 34 Nodes: 2-socket Intel Sapphire Rapids CPUs, 1TB of memory, 4 H100 GPUs per node, 6.4TB NVMe

Expanse AI Resource (NAIRR)

- New GPU computing and storage capacity recently integrated with *Expanse*
- Resources are available to the National AI Research Resource (NAIRR) Pilot project users
- Two SDSC Scalable Computing Units (SSCU) integrated by Dell with 17 GPU nodes per rack
- Each node has two 36-core Intel Sapphire Rapids CPUs, 1TB of memory, and 4 NVIDIA H100 GPUs; 6.4 TB of node-local NVMe;
- Networking uses NDR NVIDIA InfiniBand switches in the SSCUs. 2 NDR cards per node.
- NAIRR expansion will enable researchers to train complex models that can be applied to a wide range of real-life applications such as weather prediction, image classification, natural language processing, health care, software development, smart manufacturing and self driving cars.

Efficient software stack support and maintenance using modules and Spack-based installations; Containers

- *Expanse* supports a broad application base with installs and modules for commonly used packages in bioinformatics, molecular dynamics, io libraries, AI/machine learning, math libraries, quantum chemistry, structural mechanics, visualization, and others
- The bulk of the *Expanse* software stack uses Spack-based installs. Separate Spack install trees for the CPU and GPU nodes. Spack is a flexible package manager for supercomputers, the deployment tool for the Exascale Computing Project (ECP), with options to build applications and libraries from source.
- Multiple versions, compilers, platforms and configurations supported - gcc, aocc, Intel compilers, PGI (now NVHPC), and OneAPI. CPU build for AMD Rome based platform, and GPU build for Intel Xeon 6248 CPUs + NVIDIA V100, A100 GPUs.
- SDSC also builds and maintains Singularity containers. Examples include AlphaFold2, PyTorch, TensorFlow, VisIt, Paraview, Trinity, Mathematica, REDIttools, Quantum Espresso, Spark, and PRISM-PF

Getting Allocations on Expanse

- The Expanse cluster is available as an ACCESS resource. Allocations page:

<https://allocations.access-ci.org/>

- ACCESS features several levels of allocations – Explore (to get started), Discover (for modest size requests), Accelerate (large scale), and Maximize (largest option, peer reviewed). All but the Maximize category (two opportunities per year) are available all year around
- The Expanse AI Resource is currently only available via the NAIRR Pilot program. Allocations page:

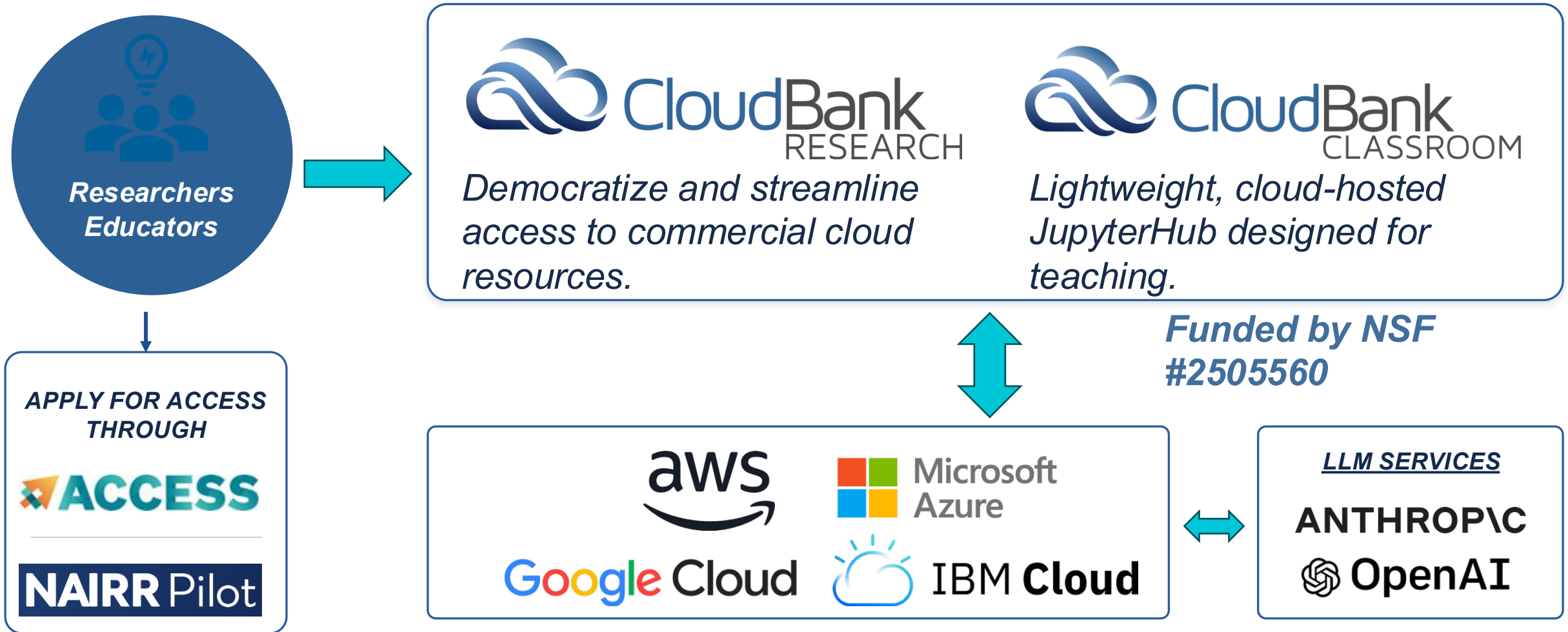
<https://nairrpilot.org/opportunities/allocations>

Cloudbank

NSF Funded Category I System

NSF Award # 2505560

CloudBank: Accelerating Science and Engineering Research in the Commercial *Cloud*



CloudBank Research

Provides direct access to multiple commercial cloud resources (managed via CloudBank portal):

- account management tools
- cost monitoring
- financial operations
- help desk support and training materials
- office hours and cloud clinics



229 ACCESS/NAIRR projects **646** cloud accounts **1094** users **Multi-cloud, pay-per-use model** (in production since Aug 2020)

Commercial Cloud Services For Research



Quantum Computing

Run quantum algorithms on real quantum hardware and simulators

Example services: AWS Braket • Azure Quantum • IBM Quantum



Foundation Models & AI

Access, fine-tune, and deploy LLMs and multimodal AI models

Example services: Amazon Bedrock • Azure AI Foundry • Gemini • Anthropic Claude • OpenAI



Edge Computing & IoT

Collect and process sensor data from distributed devices and scientific instruments

Example services: AWS IoT Core • Azure IoT Hub • Google Distributed Cloud / Edge TPU • AWS Greengrass • Azure Arc • IBM Edge Application Manager



Elastic & Distributed Computing

Verify that your tools and workflows scale smoothly from a single VM to modest multi-node and distributed setups

Example services: Amazon EC2 Auto Scaling • Google Compute • Azure Virtual Machine Scale Sets • Kubernetes (EKS/GKE/AKS)



Science Gateways & Hosted Research Services

Build web portals and managed services for research communities

Example services: AWS App Runner • Azure App Service • Google Cloud Run • Managed Kubernetes • Serverless Functions



Network Measurement

Deploy lightweight VMs worldwide to study latency, routing, DNS, Internet performance, and distributed systems

Example services: Amazon EC2 • Google Compute Engine • Azure Virtual Machines • IBM Virtual Servers

CloudBank Classroom

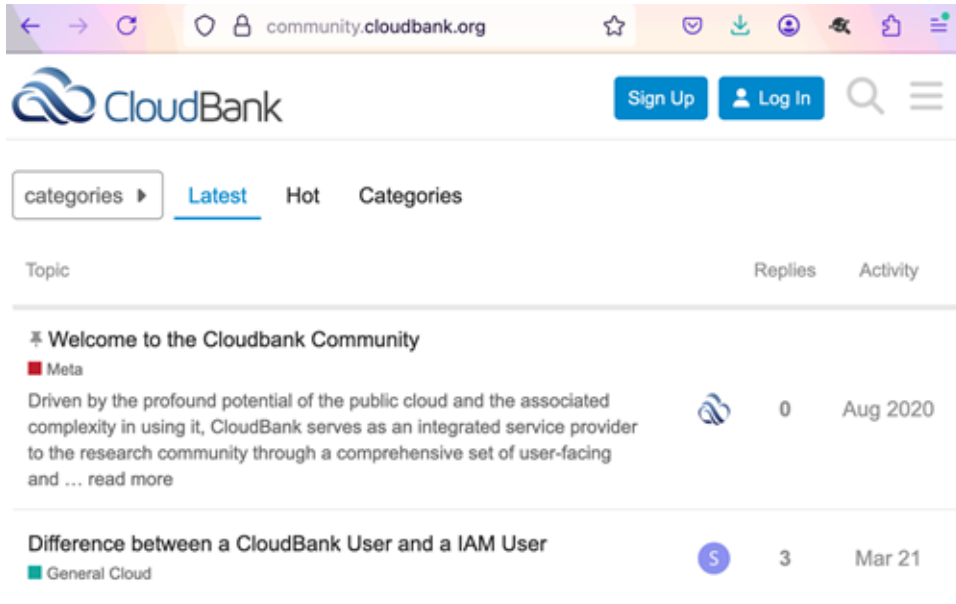
- Based on Berkeley Data Stack to provide scalable teaching both on campus and remote
- Curriculum based around interactive Jupyter notebooks; Data 8 course, etc.
- Faculty develop and deploy courses in Jupyter
- Backend: Scalable Kubernetes-based JupyterHub along with autograding and other tools
- Partnership with 2i2c.org expands reach to 23 community colleges
- *New: Teaching AI in the classroom*

49 universities **2011** students



Community Forum

<https://community.cloudbank.org>



Cloud Clinics + Office Hours

<https://www.cloudbank.org/news>

Aug 19 Noon PDT	Office Hours
TBD Cloud Clinic	TBD

Past Cloud Clinic Topics

- *Using Containers*
- *Data Publication and access*
- *Spot instances and checkpointing*

<http://www.cloudbank.org>



help@cloudbank.org

VOYAGER

NSF Funded Category II System

NSF Award # 2005369

VOYAGER

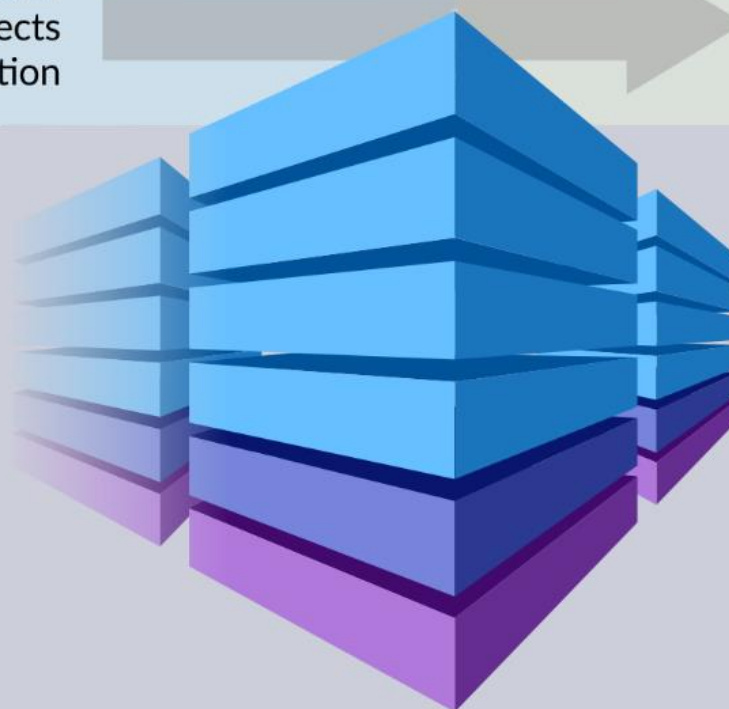
EXPLORING AI PROCESSORS in SCIENCE and ENGINEERING

3-YEAR TESTBED PHASE
Focused Select Projects
Workshops, Industry Interaction

2-YEAR ALLOCATIONS PHASE
NSF Allocations to the Broader Community
User Workshops

INNOVATIVE AI RESOURCE
Specialized Training Processors
Specialized Inference Processors
High-Performance Interconnect
X86 Standard Compute nodes
Rich Storage Hierarchy

OPTIMIZED AI SOFTWARE
Community Frameworks
Custom user-developed AI Applications
PyTorch, Tensorflow



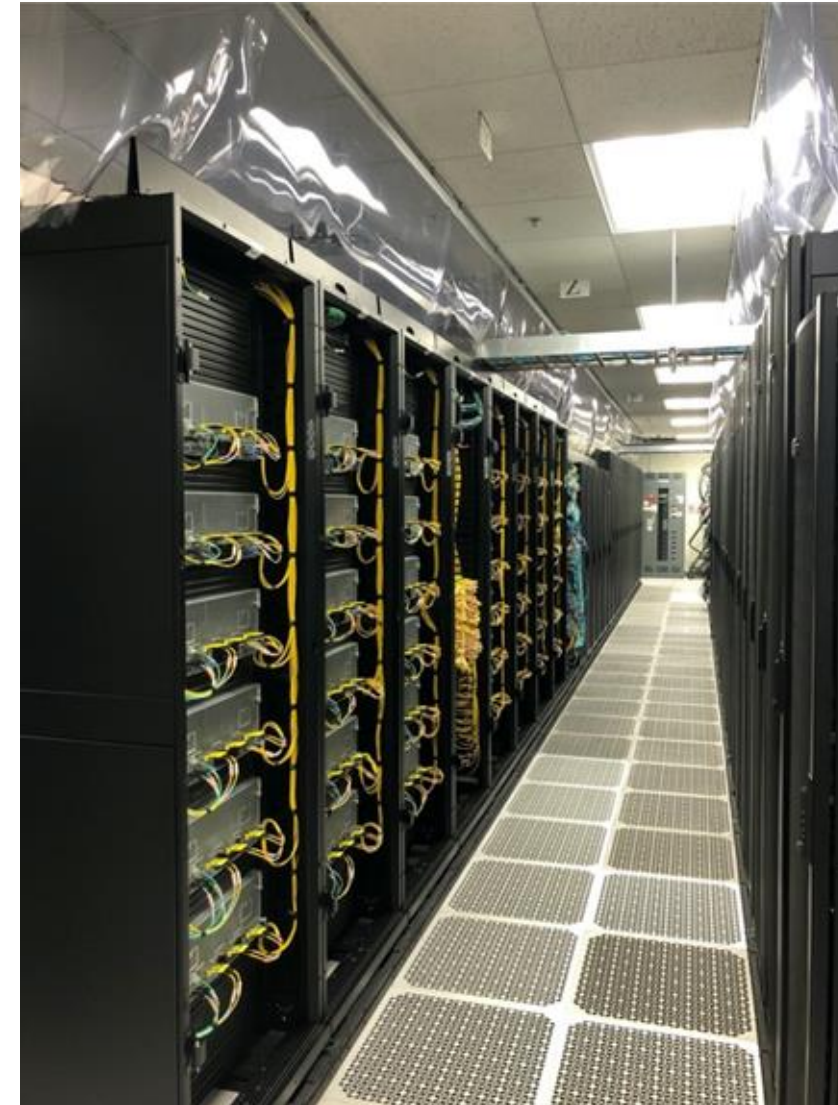
IMPACT & ENGAGEMENT
Large-Scale Models
AI Architecture Advancement
Improved Performance of AI Applications
External Advisory Board of AI & HPC Experts
Wide Science & Engineering Community
Advanced Project Support & Training
Accelerating Scientific Discovery
Industrial Engagement

Category II System, NSF Award # 2005369

PI: Amit Majumdar (SDSC); Co PIs: Rommie Amaro (UCSD), Javier Duarte (UCSD), Mai Nguyen (SDSC), Robert Sinkovits (SDSC)

Voyager: an HPC system with an innovative architecture dedicated to AI

- Envisioned as a system to facilitate exploration of a **new architectures** in support of **AI** in research and engineering.
- Its main component is the **Intel Gaudi** accelerators. They are specifically **designed** for **AI** applications, and its optimized software allows users accelerate and scale their models.
- Voyager is a NSF category II HPC system. Test phase + production.
- During the 3-year **test phase**, the Voyager team engaged with the research community. A **large number** of Deep Learning **models** from multiple fields have been deployed and analyzed on Voyager allowing us to learn more about this unique architecture.
- In the **allocation** (production) phase, Voyager opens its doors to a broader community.



Voyager is a heterogeneous system designed to support complex AI workflows

- **42x Intel Habana Gaudi** training nodes, each with 8 training processors (**336 in total**); all-to-all network between processors on a node
- Gaudi processors feature specialized hardware units for AI, HBM2, and on-chip high-speed Ethernet
- **2x first generation inference nodes**, each with 8 inference processors (**16 in total**)
- **36x Intel x86 processors compute nodes** for general purpose computing and data processing
- **400 GbE interconnect** using RDMA over Converged Ethernet
- **3 PB Storage** system connected via 25GbE. Deployed as Ceph; experiment with other filesystems
- 324 TB HFS; connectivity to compute via 25GbE
- Machine integrated by Supermicro; includes Arista switch
- ***35 of the Gaudi1 training nodes were replaced by Intel Habana Gaudi2 training nodes***

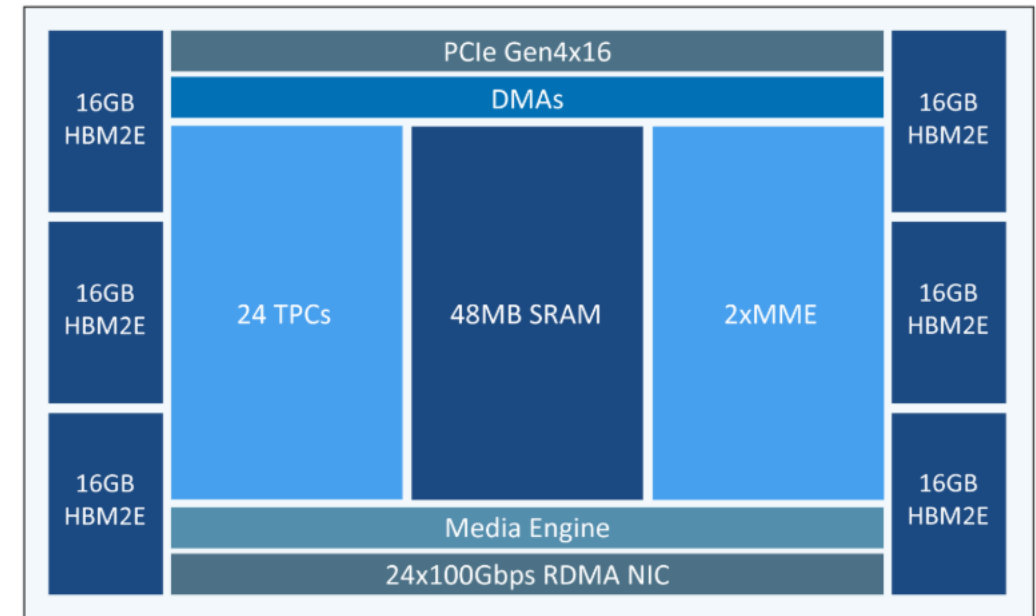
System Component	Configuration
INTEL GAUDI TRAINING NODES	
Node count	42
Training processors/node	8
Host x86 processors/node	2
Memory/node	512 GB DDR4
Memory/training processor	32 GB HBM2
Local NVMe	6.4 TB
INTEL GOYA INFERENCE NODES	
Node count	2
Inference processors/node	8
Host x86 processors/node	2
Memory/node	512 GB DDR4
Memory/inference processor	16 GB DDR4
Local NVMe	3.2 TB
STANDARD COMPUTE NODES	
Node count	36
x86 processors/node	2
Memory capacity	384 GB
Local NVMe	3.2 TB
STORAGE SYSTEM	
High performance storage: HDD:NVMe	3 PB:140 TB
High performance filesystems	Ceph, Lustre
Home filesystem storage: HDD:NVMe	324 TB: 12.4 TB
File system	NFS

Gaudi2 Architecture (35 nodes 8 HPU nodes)

- Gaudi2 node
 - 7nm process technology
 - 24 Tensor Processor Cores (TPC)
 - Dual matrix multiplication engines (MME)
 - 24 100 Gigabit Ethernet integrated on chip
 - 96 GB HBM2E memory on board
 - 48 MB SRAM
- Voyager now has 35 Gaudi2 nodes

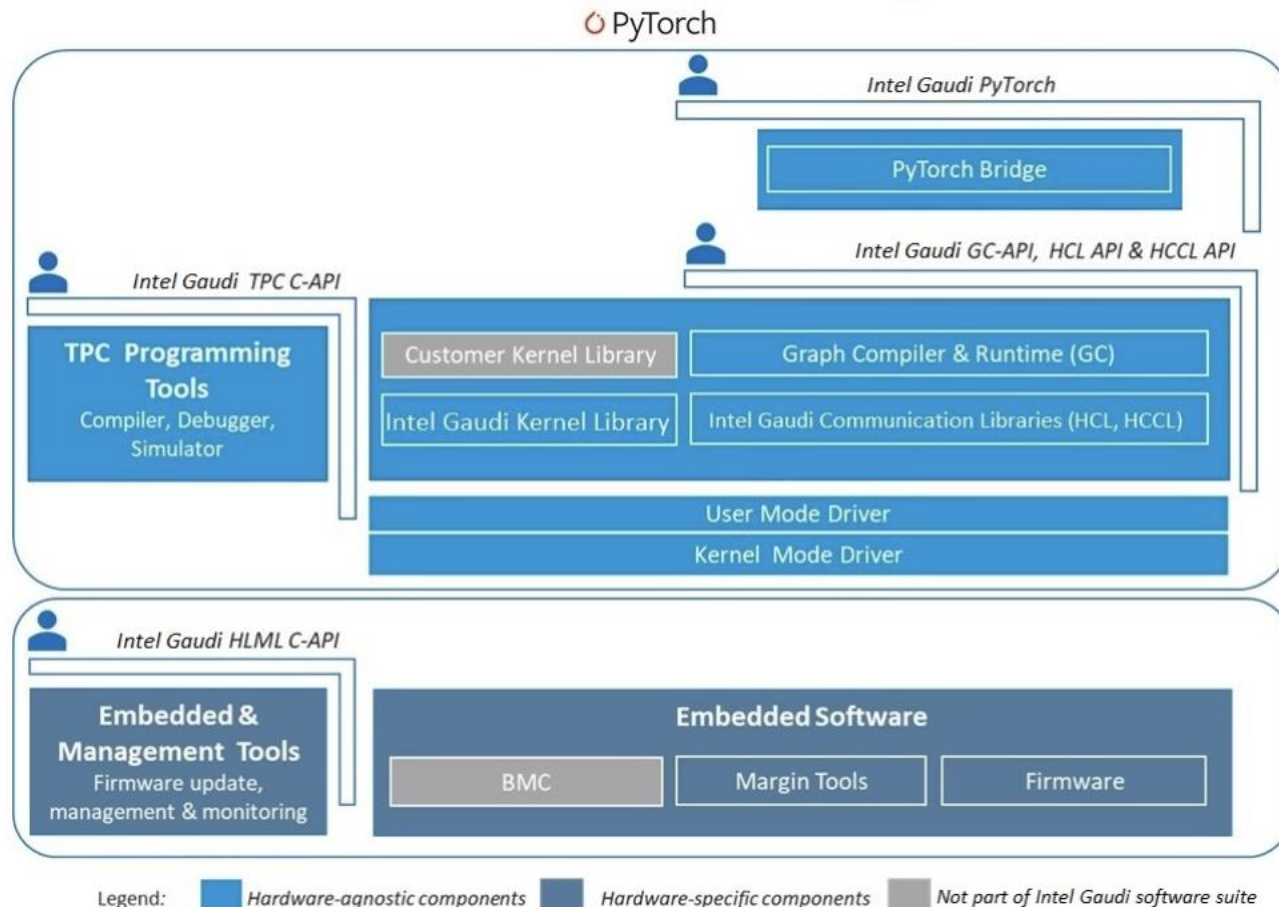
Chip Architecture Diagram

GAUDI²



https://docs.habana.ai/en/latest/Gaudi_Overview/Gaudi_Architecture.html#intel-gaudi-2-ai-accelerator

Intel Gaudi Software Suite for High Performance DL Training/Inference



Intel Gaudi Software Suite

Reference: https://docs.habana.ai/en/latest/Gaudi_Overview/Intel_Gaudi_Software_Suite.html

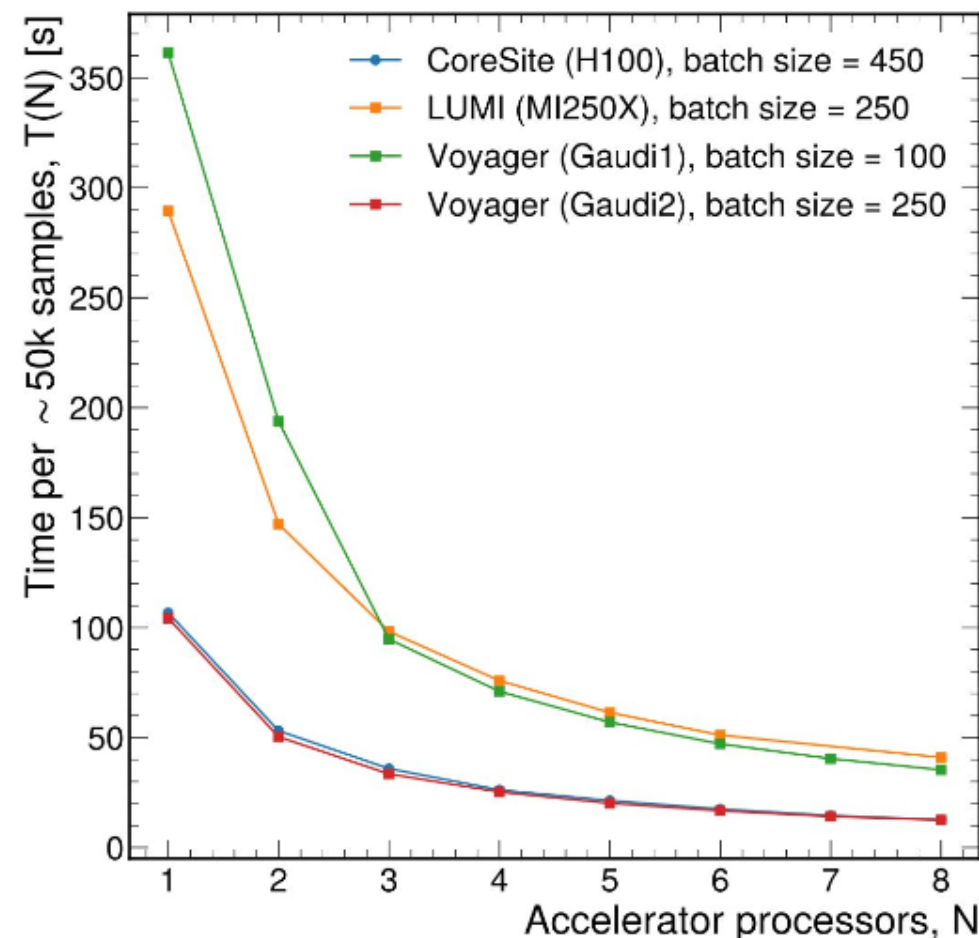
- Shared software suite for training and inference
- Start running on Habana accelerators with minimal code changes
- Integrated with PyTorch. TensorFlow supported on older Synapse versions.
- Rich library of performance-optimized kernels
- Advanced users can write their custom kernels
- Components include graph compiler and runtime, TPC kernel library, firmware and drivers, and developer tools

Using Intel Gaudi Accelerators

- At a high level, users need little code modification to run their AI models
- Pytorch is natively integrated to Synapse AI (software stack) and allows to run many popular AI applications in computer vision and generative AI (Transformer, Diffusion models)
- Applications in Voyager are run in **containers** and are managed by **Kubernetes**. All jobs are run using the **kubectrl** command line interface, with yaml files to describe requirements and scripts to run
- Advanced users can write their own customized kernel
- Easy migration from GPU-based models
- **Jupyter** notebook setup is **possible** on Voyager
- To run LLMs or Diffusers from **Huggingface**, users can use **optimum-habana** for Intel Gaudi.
- Distributed training (DDP) using HCCL (equivalent of NCCL)
- Synapse AI updated regularly to enable new models, features

<https://docs.habana.ai>

Figure: MLPF scaling using different accelerators (Courtesy of Javier Duarte)



Scientific applications already running on Voyager

Field/Project	AI architecture
Cosmology: Super resolution of galactic simulations	Diffusion
Biomedical text analytics	LLM
Cardiac image analysis	U-Net
High energy physics	Graph NN
Molecular simulations	NN
Data driven weather prediction	U-Net
Human microbiome research	Categorical VAE
Dose prediction in cervical brachytherapy	U-Net
Ontology-driven finetuning of LLM for epilepsy	LLM
Research accessibility via visual representation	Diffusion

Getting Allocations on Voyager

- The Voyager category II system is now in its allocated phase and is available via both ACCESS and NAIRR Pilot programs.
- ACCESS Allocations page describes the various allocation levels and resources available:

<https://allocations.access-ci.org/>

- Voyager is also available via the NAIRR Pilot program. Allocations opportunities and application process are described below:

<https://nairrpilot.org/opportunities/allocations>

- Contact consult@sdsc.edu if you are interested in using Voyager

Useful Voyager Links

- Intel Gaudi documentation:
<https://docs.habana.ai/>
- Pytorch on Intel Gaudi:
<https://docs.habana.ai/en/latest/PyTorch/index.html>
- Reference models/Tutorials:
 - Intel-Gaudi: <https://github.com/HabanaAI/Model-References>
 - Voyager: <https://github.com/javierhndev/Voyager-Reference-Models>
- Optimum-habana:
<https://github.com/huggingface/optimum-habana>

Triton Shared Computing Cluster (TSCC)

Research Computing with Triton Shared Computing Cluster (TSCC) at SDSC



Overview

- Medium-scale **HPC** research compute cluster
- Designated as **NIST SP 800-171** compliant since Jan 2026
- Supports research with Controlled Unclassified Information (**CUI**) or Controlled Access Data (**CAD**)
- Serves a community of over **2,500 researchers** from **50+ research labs and startups**

Resources:

- High performance CPU compute nodes
- GPU-accelerated nodes for AI/ML workloads
- RoCE (RDMA over Converged Ethernet) – high-speed, low-latency network interconnect
- Scalable storage resources

Allocation Models:

- Hotel Program – pay as you go model
- Condo Program – purchase servers managed by TSCC



Information and Contact:

Website:

<https://tritoncluster.sdsc.edu>

Email:

tsc-support@ucsd.edu

Advancing Cancer Research - Mutational Signature for Interpreting and Preventing Cancer

- Studies somatic mutation and associated cancer risk by using machine learning approach to reveal the identity of the processes causing somatic mutations from genome sequencing data
- AI model to reduce the timeline for treating cancer patients by predicting the biomarker based on the initial biopsy
- With TSCC - multiple CPU/GPU servers, dedicated storage for storing and analyzing NIST 800-171 compliant cancer patient data



Alexandrov, Ludmil B et al. “Signatures of mutational processes in human cancer.” *Nature* vol. 500,7463 (2013): 415-21. doi:10.1038/nature12477

TSCC - Benefit to research group

- Professional system administrators manage purchased server, storage and provide NIST-compliant environment
- Expert user support for optimal use of the system; manage software stack; optimize code/pipeline; provide HPC/AI training.

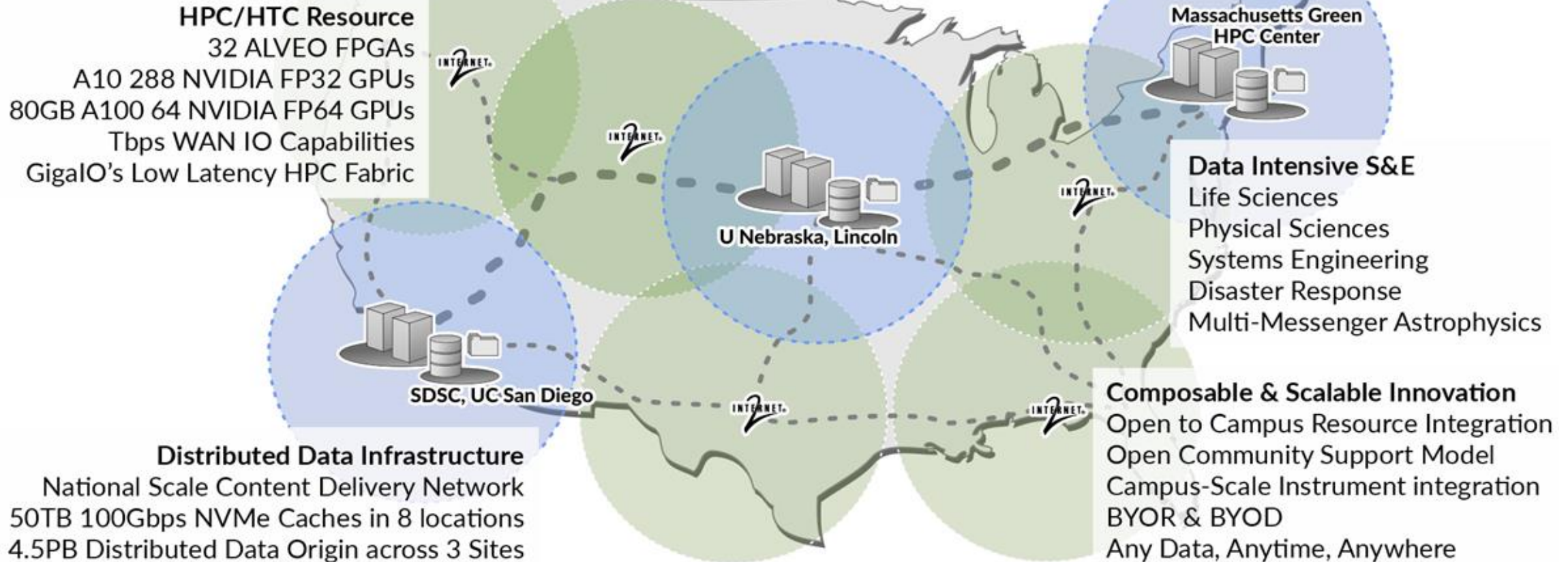
PNRP

NSF Funded Category II System

NSF Award # 2005369

NATIONAL RESEARCH PLATFORM

Designed for Growth & Inclusion



Category II: A Prototype National Research Platform (PNRP) Project (NSF OAC #2112167) added significant compute and distributed storage resources to NRP

***5-year project: \$5M for Acquisition and Deployment; \$7.25M for Operations and Maintenance
PI = Wuerthwein; Co-PIs: DeFanti, Rosing, Tatineni, Weitzel***

The National Research Platform

- The National Research Platform is a partnership of more than **125** institutions.
- Led by researchers at **UC San Diego, University of Nebraska-Lincoln, and MGHPCC.**
- Includes contributions by the **National Science Foundation**, the **Department of Energy**, the **Department of Defense**, and many research universities and R&E networking organizations in the US and around the world.
- NRP operates a primary **Kubernetes** distribution cluster with hardware spanning across **3 continents**, called **Nautilus**.

PNRP is designed & deployed as Category II innovative testbed

- NSF, Category II definition: “Resources proposed in this category will be initially deployed as a prototype/testbed system supporting S&E research through delivery of novel forward-looking capabilities and services”
 - 3-year Testbed Phase: Close collaboration with community to explore the PNRP system for transformational S&E
 - 2-year Allocation Phase: PNRP will be made available to the broader community via NSF supported allocation mechanism (e.g., ACCESS, PATH Facility, FABRIC, NAIRR, etc.)
- **Our proposal:**
 - A **nationally distributed system** with hardware deployed in 8 locations within the continental USA
 - An “**Open Cyberinfrastructure for all of Open Science**” ...
 - **Open horizontally via a Bring-Your-Own-Resources/Devices program**
 - Open vertically for higher level service deployments on our Kubernetes layer
 - A composable hardware platform to support **R&D towards domain specific architectures**
 - PCIe hardware composability => allow for nodes that combine GPUs, FPGAs, and large volumes of NVMe

Time Range **24h** 7d 30d

Updated 10:29:28 AM

SITES

129

Hosting NRP nodes



NODES

501

Registered in Kubernetes



GPUS

1.5K

Across all nodes



CPU CORES

32.3K

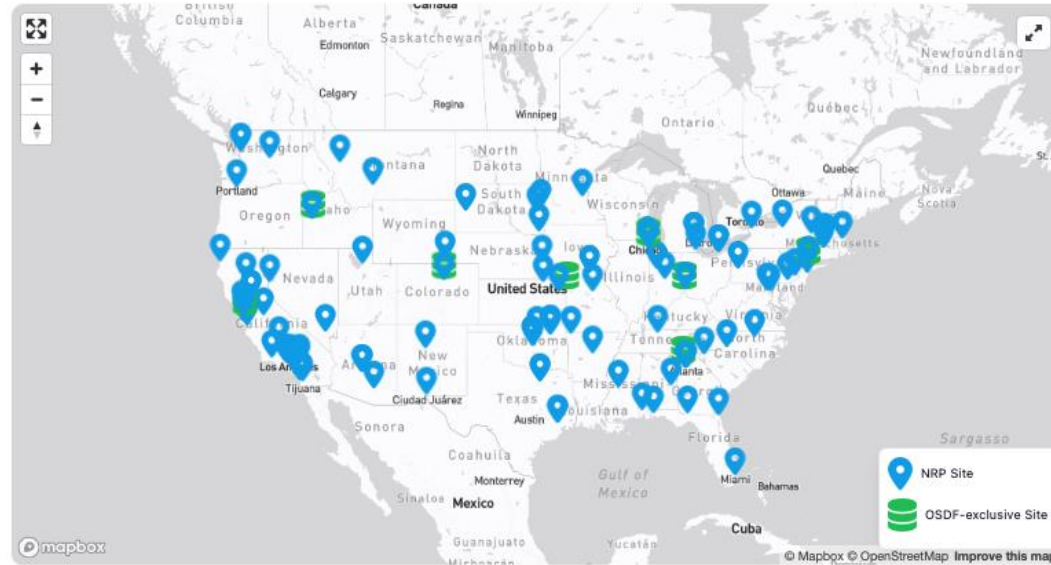
Across all nodes



OSDF NODES

31

Nodes serving OSDF cache

[View OSDF Dashboard →](#)

ABOUT NRP

NRP NATIONAL RESEARCH PLATFORM

The National Research Platform is a partnership of more than 50 institutions, led by researchers at UC San Diego, University of Nebraska-Lincoln, and Massachusetts Green High Performance Computing Center and includes contributions by the National Science Foundation, the Department of Energy, the Department of Defense, and many research universities and R&E networking organizations in the US and around the world.

SELECT SITE

Choose a site or click on the map

MAP CUSTOMIZATION

Select multiple sites to highlight

Or filter by regex pattern

GPUS ALLOCATED

1,467

→ 0%

May 3, 2026 at 10 AM



May 2, 2026 at 10 AM

May 3, 2026 at 10 AM

RUNNING JUPYTER PODS

111

↗ 11%

May 3, 2026 at 10 AM



May 2, 2026 at 10 AM

May 3, 2026 at 10 AM

ACTIVE RESEARCH GROUPS USING GPUS

95

↘ 2%

May 3, 2026 at 10 AM



May 2, 2026 at 10 AM

May 3, 2026 at 10 AM



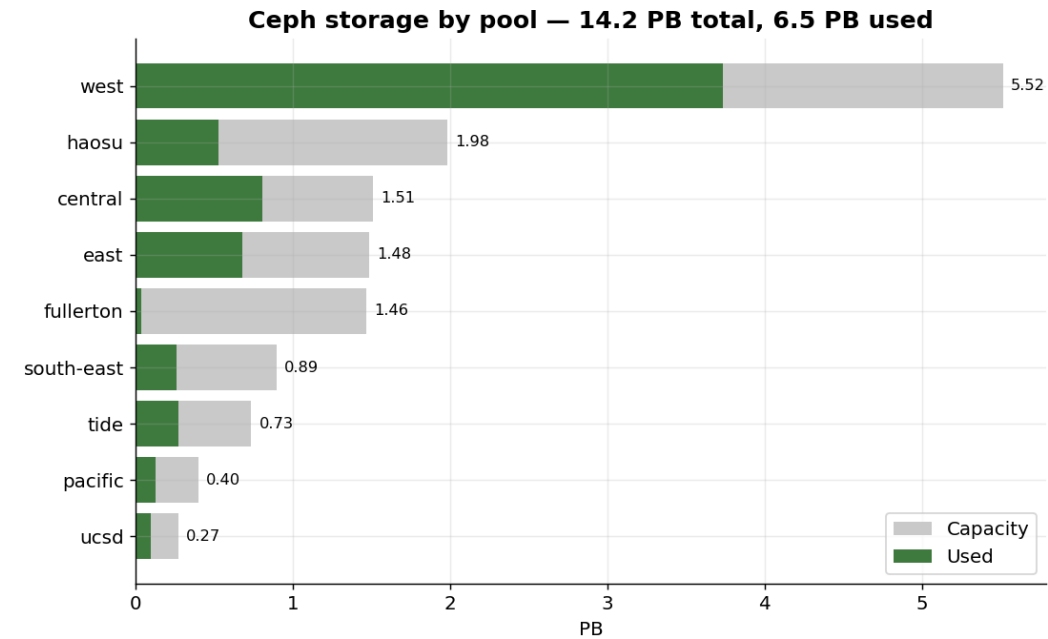
This work was supported in part by National Science Foundation (NSF) awards CNS-1730158, ACI-1540112, ACI-1541349, OAC-1826967, OAC-2112167, CNS-2100237, CNS-2120019.

<https://dash.nrp-nautilus.io/>

BYOR Program — deployed hardware, July 2026

336 GPUs hosted at non-R1 colleges/universities · 276 in EPSCoR states*

Accelerator	# in NRP	Accelerator	# in NRP
A10	279	RTX 4090	24
RTX 3090	212	A40	12
A100 80G (SXM)	127	L40S	12
RTX 2080 Ti	113	A100 40G	9
V100	106	RTX PRO 6000 Blackwell	8
L4	96	H100 80G	8
GTX 1080 Ti	86	GTX 1080	8
Other (nodes down/unlabeled)	82	RTX 5000 Ada	8
L40	68	RTX A5000	6
RTX A6000	60	H200 NVL	6
A100 80G (PCIe)	36	Other	5
RTX A4000	32	RTX 4000 Ada	2
Titan / Quadro	28	GH200 480G	1
Total		1434	

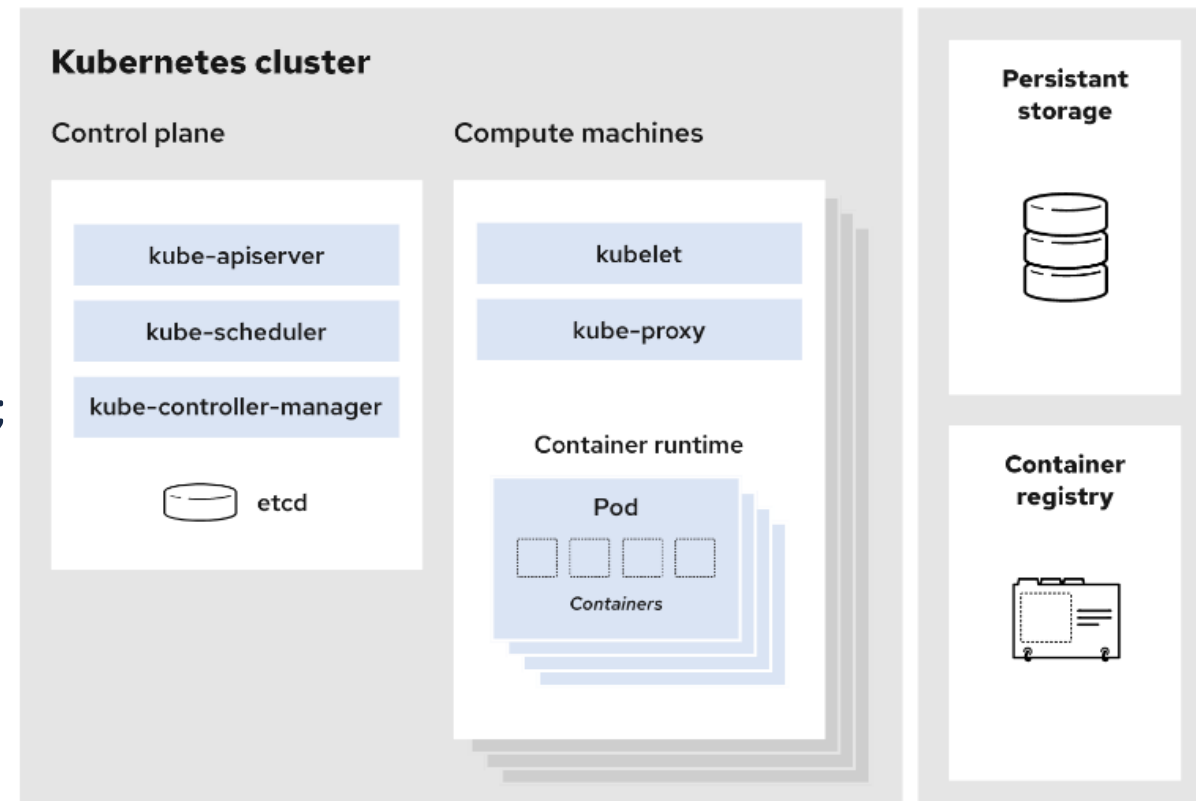


In addition: 32 Xilinx U55C FPGAs. GPU counts are installed hardware (kubectl inventory, MIG-corrected); online counts run ~5–7% lower as nodes cycle.

**non-R1 per Carnegie 2025 (US academic, not R1); EPSCoR overlaps non-R1. Cross-check: dash.nrp-nautilus.io reports 1,340 GPUs / 503 nodes / 128 sites (different MIG counting, 23 nodes missing there). Source: cluster node inventory + node→institution mapping, 2026-07-10*

The Nautilus Cluster

- Nautilus is a Kubernetes cluster
- Control plane: Manages the cluster and the worker nodes
- Worker Nodes: Maintain running pods and provide Kubernetes runtime environment
- Has been in continuous operation for **6 years**.
- Various storage options like **CephFS; CvmFS; S3**; advanced monitoring tools like **PerfSONAR; traceroute; Prometheus**.
- Diverse computation and data tools: **JupyterHub; WebODM; GitLab; Nextcloud; Overleaf**.
- Collaboration and communication: **Jitsi; EtherPad; Hedgedoc; SyncThing**.
- **500+ nodes; 1500+ GPUs; 50+ FPGAs; Bluefield 2 SmartNICs; SmartSSDs; programmable Tofino switches....**



Hardware on Nautilus

The cluster has different types of resources including CPU, GPU, and FPGAs

- **CPU**s: 16-384 cores/node
- **GPU**s: max 16 in a node
- **FPGA**s: max 4 in a node
- **Memory (RAM)**: 16GB – 2.16 TB/node
- **Disk Space (local scratch)**: 77GB – 20TB /node

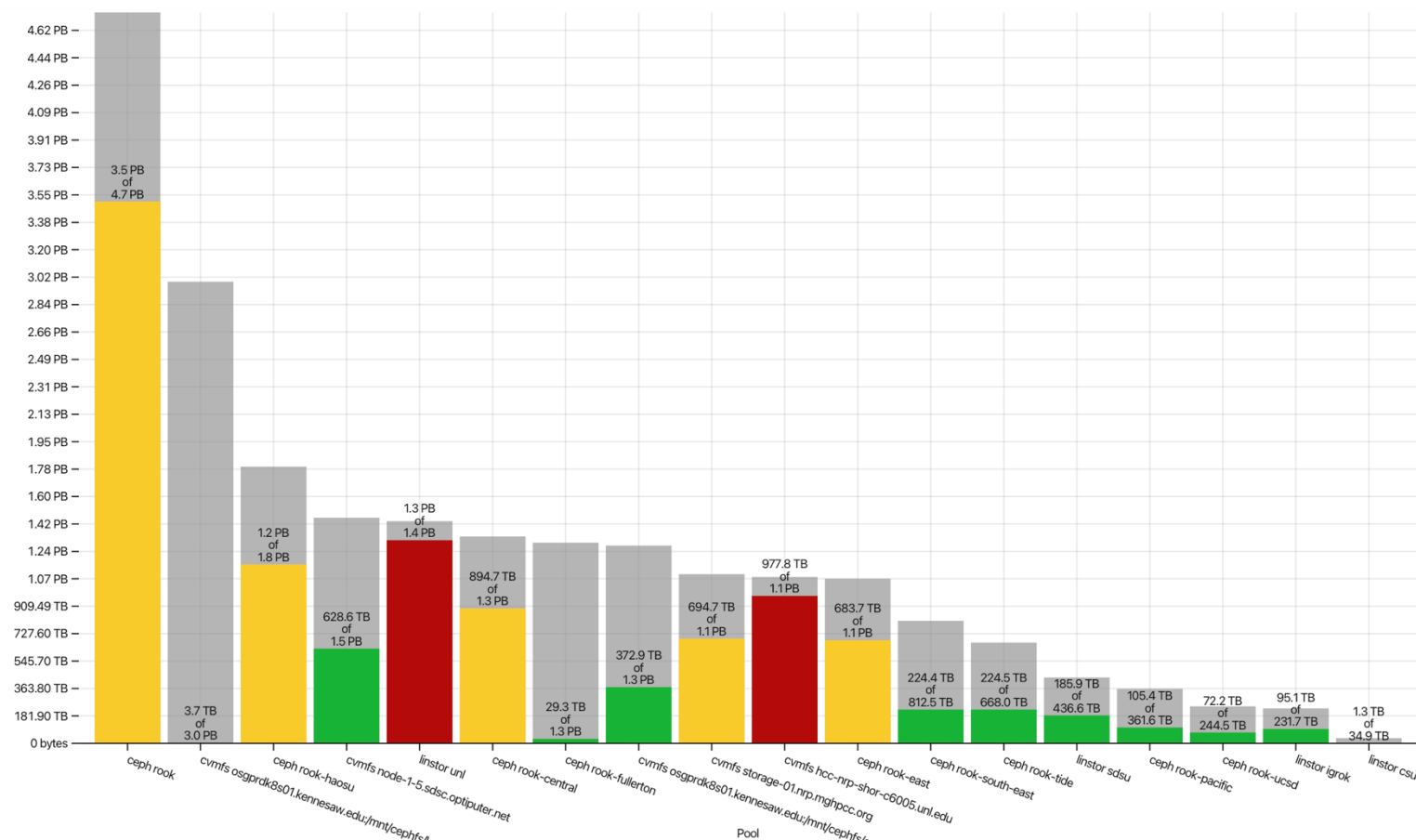
- ConnectX-6 NICs, P4-programmable **Tofino** Switches
- **32** P4-programmable Xilinx Alveo U55C **FPGAs**
- **16** BitWare 520N-MX Stratix **FPGAs**
- **3** Xilinx Alveo SN1000 **FPGAs**
- **7** NVIDIA Bluefield 2 **DPU**s
- **2** Xilinx U.2 **FPGA**-powered SmartSSDs
- Kubernetes cluster with *Multus*, *Calico* and *ESnet SENSE* support

<https://nrp.ai/viz/resources>

Storage Options on Nautilus

- Ceph File System → Ceph FS → ReadWriteMany PersistentVolumes
- Ceph Rados Block Device → Ceph RBD → ReadWriteOnly PersistentVolumes
- Linstor Block → ReadWriteOnly PersistentVolumes
- S3 Compatible storage → Bucket Endpoints
- Open Science Data Federation → OSDF

<https://nrp.ai/viz/storageall/>



Classroom Support

- **Interactive access via Jupyter is a foundational feature of PNRP**
 - > 1000 users from 47 institutions used our JupyterHub
 - >100 namespaces have deployed their own JupyterHub instances
 - **Broader Impact & Open CI Vision & Jupyter → Education as a focus on PNRP**
- **NAIRR Classroom offering – over 25 allocations made on PNRP**
 - Students can login using CILogon with institutional credentials
 - Professor and TA will have admin status to create namespaces for classes
 - GPU nodes each with 8 A10 GPUs, 512GB of RAM, 2 AMD EPYC 7502 CPUs, and 8TB of NVMe
 - Support/Community interaction through Matrix channels
 - Direct ticketing system available for any issue needing longer follow ups
 - <https://nairrpilot.org/opportunities/education-call>

Accessing Nautilus and NRP



JupyterHub



Coder



Kubernetes

NRP hosted JupyterHub and Coder services available. Also provide templates for custom deployments of both options

Command line interface using kubectl

Useful NRP Links

- Documentation:

<https://nrp.ai/>

- Training resources:

<https://training.nrp-nautilus.io>

- NRP-managed LLMs:

<https://nrp.ai/documentation/userdocs/ai/llm-managed/>

- NRP hosted JupyterHub:

<https://nrp.ai/documentation/userdocs/jupyter/jupyterhub-service/>

- NRP Storage Options:

<https://nrp.ai/documentation/userdocs/storage/intro/>

COSMOS

NSF Funded Category II System

NSF Award # 2404323

COSMOS

DEMOCRATIZING the ACCELERATOR ECOSYSTEM
for SCIENCE and DISCOVERY

*A category II testbed system
supported by NSF Award #
2404323, Office of Advanced
Cyberinfrastructure (OAC).*

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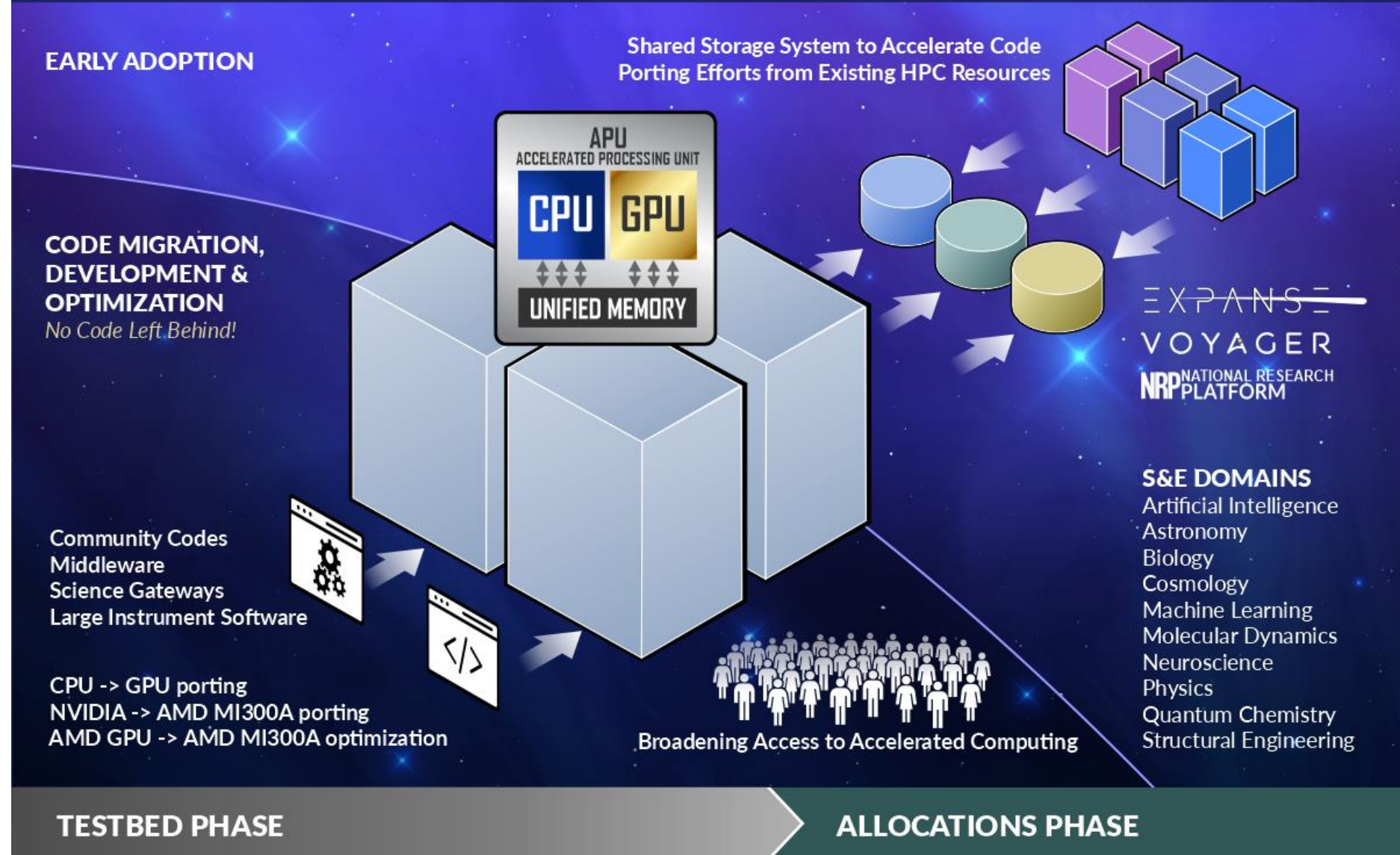
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Christopher Irving

Senior Personnel:

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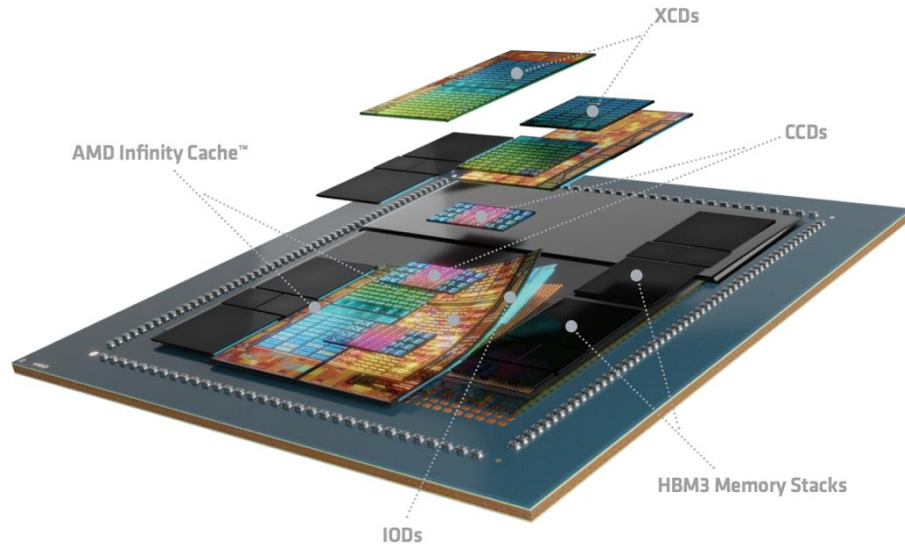
Mary Thomas



Cosmos: Democratizing the Accelerator Ecosystem for Science and Discovery

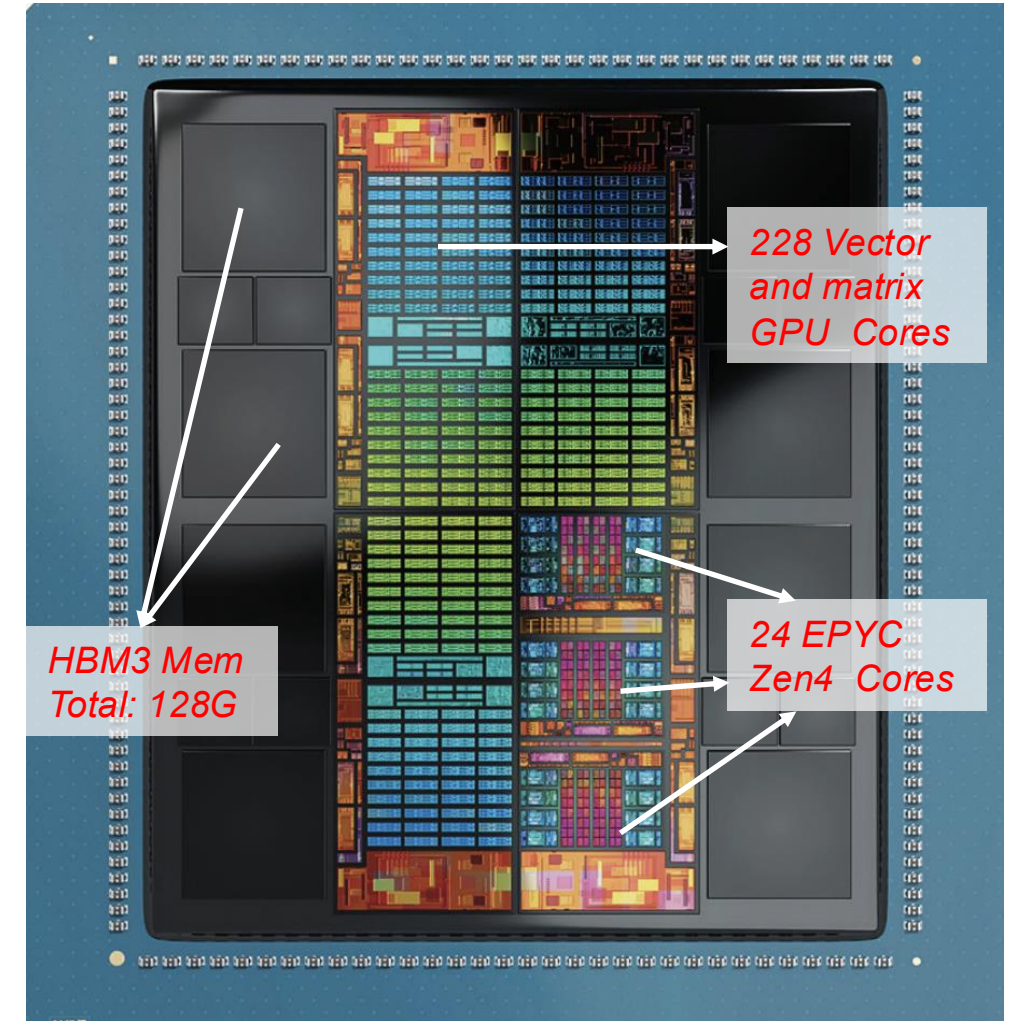
- Ease the pathway for accelerating a wide range of AI, science and engineering applications, including many that have not yet been ported to GPUs.
- Enabling technology is the AMD Instinct MI300A Accelerated Processing Unit (APU) that integrates CPU and GPU capabilities with unified memory.
- APU memory architecture facilitates an incremental programming approach, easing porting of applications, and enabling many communities to adopt GPU acceleration on this first NSF datacenter APU-based HPC system.
- HPE Cray Supercomputing EX2500 with HPE Slingshot interconnect, 4-socket nodes with all-to-all connectivity using AMD high-speed interconnect. Fully liquid cooled.
- Flash based storage for high IOPs and bandwidth supporting anticipated application workload.
- Research collaborations with applications from artificial intelligence, astronomy, neuroscience, molecular biology, structural engineering and more. Target community codes, science gateways and enabling middleware.

AMD MI300A Architecture



- MI300A APU combines CPU, GPU, and memory on one package. Six accelerated compute dies (XCDs) combined with 3 chiplets with 8 zen4 cores each.
- 128 GB of HBM3 memory shared coherently between CPUs and GPUs. 5.3 TB/s peak throughput.
- 256 MB AMD Infinity Cache (last level) shared between XCDs and CPUs
- Design helps eliminate need to do host/device data copies and eases code development (*going towards our goal of democratizing access to accelerators!*)

Reference: <https://www.amd.com/content/dam/amd/en/documents/instinct-tech-docs/white-papers/amd-cdna-3-white-paper.pdf>



HPE CRAY EX2500 Supercomputer



Single rack of EX2500



Compute Chassis: Provides power, cooling, system control, and network fabric for up to 8 compute blade slots

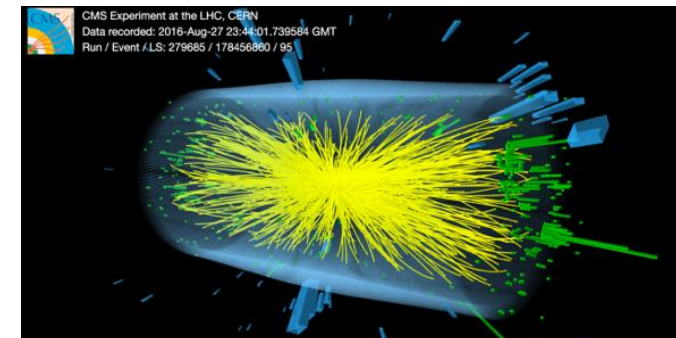
3 Chassis per rack, 8 blades per chassis, and each EX255a blade can support:

- *Two 4-socket AMD MI300a Accelerator APU nodes*
- *128GB HBM3 per APU*
- *Up to 8 HPE Slingshot 200Gbit/sec ports per blade*
- *0 or 1 local NVMe M.2 SSD per node (up to 2 per blade)*
- *2 Board Management Controllers (BMC) per blade*
- *Cooled with cold plate*

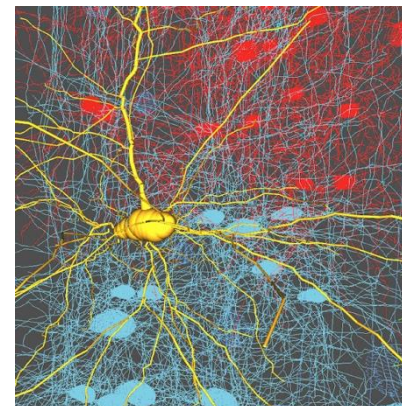
Reference: <https://www.hpe.com/psnow/doc/a00094635enw>

Cosmos will support a wide range of applications

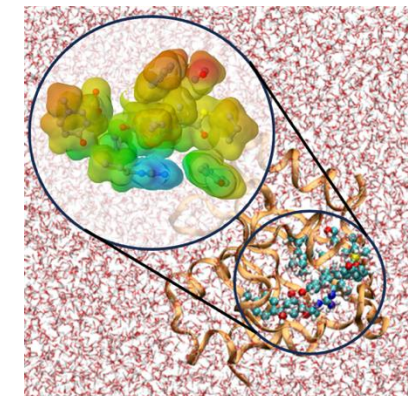
- APU memory architecture will remove need for host/device memory management that often makes porting to accelerators difficult
- The shared memory between CPUs and GPUs will allow for an incremental approach to porting new codes and provide an easier programming path
=> no code left behind!
- Contacted several research groups and communities during proposal development who will collaborate on porting and optimization efforts during testbed phase. Spans several domain science and AI/ML application areas including:
 - Heavily used community codes, such as AMBER (molecular dynamics), Enzo (cosmology), and NEURON (neuroscience);
 - Science gateways, e.g. I-TASSER and the Neuroscience Gateway;
 - Middleware, such as HTCondor and MVAPICH;
 - Software to analyze data from large instruments, such as LIGO and IceCube.



Collision readout from CMS at LHC



3D representation of M1 cortical network, output from NEURON



QM electron density at the protein active, AMBER/Quick Code

Getting Allocations on Cosmos

- The Cosmos category II system is currently in its initial 3-year testbed phase
- Contact us at consult@sdsc.edu if you have a project that might be a good fit for the system in the testbed phase
- SDSC staff will follow up and collect more information on your project and evaluate if the proposed work is a good fit



Questions

- Contact SDSC HPC User Support consult@sdsc.edu
- Visit the SDSC website: www.sdsc.edu